Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/25/2005
PI Name:	Koplik, Joel Ph.D.		
Project Title:	Molecular Dynamics of Fluid-Solid Systems		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10031	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	11/01/2003	**End Date:**	11/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2007 per PI (10/06) Note - grant renewed for 4 yrs/new NAG#		
Key Personnel Changes/Previous PI:	Koplik, Joel		
COI Name (Institution):	Banavar, Jayanth (Pennsylvania State University)		
Grant/Contract No.:	NNC04GA27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The numerical calculations are based on Molecular Dynamics, wherein one integrates the (classical) equations of motion for interacting molecules of various species, based on a prescribed interaction and specified thermodynamic operating conditions and mechanical forces. Within the framework of molecular dynamics, it is straightforward to impose a desired value of gravity or simulate the presence of g-jitter. The objective of this theoretical research is to examine the molecular behavior of fluids in confined geometries, to determine flow boundary conditions, and to investigate non-Newtonian rheology, using molecular dynamics calculations and the statistical mechanics of classical fluids. We consider temporal and spatial scales where continuum fluid mechanics provides little insight. Examples under investigation include (1) phase behavior and spreading of surfactant solutions, (2) solidification, evaporation, coalescence and rupture of non-Newtonian fluid drops and filaments, (3) boundary conditions for dilute gases and other interfaces (4) dynamics of a polymer molecule in solution, and (5) the physics of colloidal hard sphere systems. We intend to carry out a number of calculations involving coexisting liquid, vapor and solid phases under flow conditions, with two common features. First, the systems will be studied at the molecular scale to obtain otherwise unavailable information. Second, the problems to be studied deal with fundamental issues in the physics of fluids and interfaces that are of relevance to the microgravity program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	1. In order to investigate the question of possible slip at a liquid-liquid interface, we have undertaken molecular dynamics simulations of the Couette and Poiseuille flows of two-layered liquid systems with various molecular structures and interactions. In all cases, the average velocity is found to vary continuously across the interface. However, when the total liquid density in the interfacial region drops significantly compared to the bulk values, the tangential velocity varies very rapidly there, and would appear discontinuous at a coarse or continuum resolution. The value of this apparent slip varies linearly with the shear stress at the interface, with a constant of proportionality depending on the nature of both liquids but not the flow configuration. Thus, a version of the Navier boundary condition appears to apply in this situation. 2. We are investigating the flow of a nano-scale incompressible liquids confined to "chemical channels" on a substrate: a region of solid which is completely wetting to a particular liquid, surrounded by non-wetting solid regions. For the case of a long straight wetting strip, the a uniform strip of liquid is observed in molecular dynamics simulations to develop bulges or pearls, due to a surface-tension instability. The pearls propagate when the liquid moves under a pressure gradient, and combine in pairs due to non-linear wave interactions, but do not obstruct transport. A stability calculation and a simple long-wavelength approximation give qualitatively consistent results, and a paper summarizing this work is in preparation. We have continued with a study of "junction" flows, where the chemical channel has diverging and converging elements. Again, liquid moves just along the channel in the presence of a pressure gradient, and a pearling instability appears but does not obstruct transport. We are now focusing on issues of how to control switching and bifurcation effects at junctions. 3. Physics has provided an understanding of the essential features underlying the phases of inanimate matter in terms of the principles of geometry and symmetry. Living matter is also governed by physical law. The hallmarks of life are information, replication, natural selection, and functionality. The DNA molecule is the repository of information. Base-pairing and the double helix structure of the DNA molecule provides a logical mechanism for replication. A similar simple understanding of the physics of protein molecules has been lacking. We have recently shown that a flexible tube subject to compaction displays a remarkable array of protein-like properties. The constructive hypothesis that this striking resemblance is not accidental leads not only to a unified framework for understanding protein folding, amyloid formation and protein interactions but also has implications for natural selection.
Bibliography Type:	Description: (Last Updated: 12/21/2007)
Articles in Peer-reviewed Journals	Drazer G, Khusid B, Koplik J, Acrivos A. "Hysteresis, force oscillations, and nonequilibrium effects in the adhesion of spherical nanoparticles to atomically smooth surfaces." Phys Rev Lett. 2005 Jul 1;95(1):016102. Epub 2005 Jun 30. PMID: 16090633 , Jul-2005
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Articles in Peer-reviewed Journals	Hoang TX, Trovato A, Seno F, Banavar JR, Maritan A. "Geometry and symmetry presculpt the free-energy landscape of proteins." Proc Natl Acad Sci U S A. 2004 May 25;101(21):7960-4. Epub 2004 May 17. PMID: 15148372 , May-2004
Articles in Peer-reviewed Journals	Banavar JR, Hoang TX, Maritan A, Seno F, Trovato A. "Unified perspective on proteins: a physics approach." Phys Rev E Stat Nonlin Soft Matter Phys. 2004 Oct;70(4 Pt 1):041905. Epub 2004 Oct 27. PMID: 15600433 , Oct-2004
Articles in Peer-reviewed Journals	Banavar JR, Cieplak M, Maritan A. "Lattice tube model of proteins." Phys Rev Lett. 2004 Dec 3;93(23):238101. Epub 2004 Nov 29. PMID: 15601206 , Dec-2004